Getting Started with DAX

30-Minute Hands-On Exercise | Week 9 - Section 4B

Learning Objectives: Distinguish implicit from explicit DAX calculations | Write correctly structured DAX expressions using proper syntax | Classify a DAX expression as a Measure or a Calculated Column | Identify common DAX aggregation functions and when to use them | Apply DAX concepts to real analyst scenarios

Instructions: Complete all four parts in 30 minutes. Refer to your Section 4B notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 4B.

1.	When a numeric field is dragged into a visualization's data well, Power BI automatically creates an ________________________ , a DAX calculation applied behind the scenes without any code being written by the analyst.
2.	The default aggregation used by an implicit DAX measure is ________________________ , though it can be changed by clicking to select a different option.
3.	All DAX calculations -- measures, calculated columns, and calculated tables -- are written in the ________________________ in Power BI Desktop.
4.	In a DAX expression, column names and measure names must always be wrapped in ________________________ .
5.	When a table name contains a space, it must be wrapped in ________________________ inside a DAX expression.
6.	A DAX expression that cycles through each row of a table, stores a computed value per record, and adds it as a new column in the data model is called a ________________________ .

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term.

Term / Concept
A. Implicit Measure

Description / Example
1. Returns the arithmetic mean of all non-blank values in a numeric column

B. Explicit Measure	2. A named DAX calculation saved to the model that can be referenced by other measures and used in visuals
C. Calculated Column	3. Returns the total number of rows in a specified table
D. SUM()	4. A DAX expression that adds a new column to a table with row-level computed values stored in the model
E. COUNTROWS()	5. A DAX calculation Power BI creates automatically when a numeric field is added to a visual; no formula visible
F. AVERAGE()	6. Adds up all values in a numeric column and returns the total

Part 3: DAX Syntax Analysis (~8 minutes)

For each DAX expression below, identify whether it is a Measure or a Calculated Column, then describe in plain English what it calculates. All expressions use a table named 'order detail'.

#	DAX Expression	Measure or Calc. Column?	What does it calculate?
1	`Sales Total = SUM('order detail'[Amount])`		
2	`# of Orders = COUNTROWS('order detail')`		
3	`Avg Order Value = [Sales Total] / [# of Orders]`		
4	`Sales Tax (7%) = 'order detail'[Amount] * .07`		
5	`Annual Sales Goal = 1000000`		
6	`Monthly Sales Goal = [Annual Sales Goal] / 12`		

Hint: A Measure returns a single summarized value and is calculated at query time. A Calculated Column cycles through each row and stores a value per record. Look for a summarization function (SUM, COUNT, AVERAGE, etc.) to help identify measures.

Part 4: Short Answer (~9 minutes)

Answer each question in 2-4 sentences in your own words.

Question 1

A report shows a bar chart with 'Sum of Amount' as the Y-axis label. A stakeholder wants to rename this to 'Total Sales' and use it in three other visuals on the report. Why is the current calculation not ideal for this, and what should the analyst create instead?

Question 2

An analyst creates the measure: [Sales Total] = SUM('order detail'[Amount]) and then creates a second measure: [Avg Order Value] = [Sales Total] / [# of Orders]. Explain why being able to reference one measure inside another is useful. What problem would arise if measures could not reference each other?

Question 3

A team member wants to count the number of orders in the 'order detail' table. They debate between writing a Calculated Column that puts a '1' in every row or writing a Measure using COUNTROWS(). Which approach is correct for this use case, and why?

Question 4

An analyst tries to write a Calculated Column expression that looks like this: Sales Tax = SUM('order detail'[Amount]) * .07. Their instructor says this is the wrong approach. Why is SUM() incorrect in a Calculated Column, and what is the correct expression to use instead?

Bonus / Reflection (if time allows)

Section 4B introduces five common aggregation functions -- SUM, COUNT/COUNTA/COUNTROWS, AVERAGE, MIN, and MAX -- and notes that DAX also includes many other advanced functions such as CALCULATE, FILTER, ALL, RELATED, and USERELATIONSHIP. In 3-5 sentences, describe a real reporting scenario where a simple SUM() measure would not be enough and one of the advanced functions would be needed. What does the advanced function do that SUM alone cannot?
